# Superpolynomial online mistakes for isotropic Bayesian ReLU networks

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**Abstract**

We construct a fixed order of distinct Boolean inputs on which exact Bayesian prediction with an ordinary one-hidden-layer ReLU network makes superpolynomially many mistakes in the strong-prior limit. The weights have independent Gaussian coordinates, with identical variances within each layer. Before observing labels, the prior is conditioned on an absolute output margin at every input; training then requires exact fit. The construction works at every sufficiently large finite hidden width. For a parameter $m$, the input dimension is $O(m^{11})$, the target has an $O(m^{11})$-gate Boolean circuit and a shallow ReLU teacher of width $O(m^{12})$ whose coefficients are bounded by that width, while $2^{\lfloor (m-1)/2 \rfloor}$ designated predictions have correct-label posterior probability tending to zero. Thus the lower bound is stretched exponential in these explicit teacher resources. The width threshold and the calibration prefix are large, and the input order is deliberately unrepresentative of the uniform input distribution. The theorem concerns cumulative online mistakes, not iid sample complexity.

The proof determines an exact weighted Fourier norm for ReLU functions on an unequal-rail parity code, proves a polynomial stability estimate, and uses a weighted calibration argument to exclude all off-code competitors. A Bernstein representation of the sharp ReLU parity cost orders exponentially many competing completions strictly. The resulting energy gaps apply to the full Gaussian posterior, including all parameterizations of each function.

## 1 Model and main result

A target can have a small circuit and still be learned poorly along an unfavorable sequence of inputs. The issue for a Bayesian neural network is whether its actual posterior can repeatedly prefer a wrong completion, even when the learner is arbitrarily wider than a small teacher. We prove that this can occur with ordinary ReLU activation and isotropic Gaussian coordinates. No rule for selecting among tied minimum-energy classifiers is used.

The input space is $X = \{-1, 1\}^d$. Write $\bar{x} = (1, x)$ and consider

$$F_\theta(x) = \sum_{j=1}^{N} v_j \,\mathrm{ReLU}(u_j \cdot \bar{x}), \qquad \mathrm{ReLU}(t) = \max(t, 0), \quad u_j \in \mathbb{R}^{d+1}. \tag{1}$$

The first coordinate of $u_j$ is the weight of the constant input. There are no additional biases. Independently,

$$(u_j)_i \sim \mathcal{N}(0, N\sigma^2/d), \qquad v_j \sim \mathcal{N}(0, \sigma^2/N). \tag{2}$$

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Its density is proportional to $\exp[-K_N(\theta)/(2\sigma^2)]$, where

$$K_N(\theta) = \frac{d}{N}\sum_j \|u_j\|_2^2 + N\sum_j v_j^2. \tag{3}$$

The prior is first conditioned on $|F_\theta(x)| > 1/2$ at every $x \in X$. After each prediction it is conditioned on the exact revealed label. Thus a history $H \subseteq X$ with target $f : X \to \{-1, 1\}$ conditions the Gaussian law on

$$|F_\theta(x)| > 1/2 \quad (x \in X), \qquad f(x)F_\theta(x) > 1/2 \quad (x \in H). \tag{4}$$

The strong-prior limit is $\sigma \downarrow 0$ at each fixed $d, N$.

The conclusion also holds for any two fixed positive layer variances $\alpha\sigma^2$ and $\beta\sigma^2$ in place of (2). Positive homogeneity gives the lower bound $2V(F)/\sqrt{\alpha\beta}$, where $V$ is the variation norm defined below. Balancing any finite representation of cost $B$ attains energy $2B/\sqrt{\alpha\beta}$. The constructed minimizers attain this bound at the widths used below, so their strict energy order is unchanged. In particular, writing $t = \sqrt{N}\,\sigma$ turns (2) exactly into first-layer variance $t^2/d$ and readout variance $t^2/N^2$. At each fixed width the limits $t \downarrow 0$ and $\sigma \downarrow 0$ are the same.

**Theorem 1** (A fixed sequence with many confident errors)**.** *For every integer $m \geq 32$, set*

$$R = 2^{60}m^{10}, \quad \delta_i = 2(2m + 2i - 1)\ (1 \leq i \leq m), \quad n_{\max} = mR + 6m^2,$$

$$d = 2mR + 6m^2, \qquad N_0 = 2^{m+n_{\max}+1}.$$

*There are a Boolean target $f_m$ on $d$ bits, a fixed permutation of its input cube, and*

$$T_m = 2^{m-\lfloor m/2\rfloor-1} = 2^{\lfloor (m-1)/2\rfloor}$$

*designated rounds such that, for every finite $N \geq N_0$, the posterior probability of the correct label at each designated round tends to zero as $\sigma \downarrow 0$.*

*The target has a bounded-fan-in Boolean circuit with $O(d)$ gates and a one-hidden-layer ReLU implementation of width at most*

$$C_m = (m + 1)(n_{\max} + 1) = O(m^{12}),$$

*output margin one, and every coefficient bounded by $C_m$. The same input permutation works for every $N \geq N_0$.*

*For every $\eta > 0$ and fixed such $m, N$, all sufficiently small positive $\sigma$ therefore have these properties:*

1. *Fresh posterior sampling makes at least $T_m$ mistakes with probability at least $1 - \eta$.*
2. *Posterior-label majority makes a mistake at every designated round.*
3. *A prediction rule requiring correct-label posterior probability strictly greater than $2/3$ fails at every designated round.*

The width threshold is an explicit sufficient bound, not a sharp one. The theorem proves super-polynomial cumulative mistakes in the displayed teacher resources; it does not assert $\exp(\Omega(C_m))$ mistakes or cover every width merely exceeding $C_m$. Its calibration prefix reveals all inputs outside a small union of logical fibers. This is compatible with good iid learning guarantees.

## 2 Variation norm and parity codes

All Fourier coefficients and $L^2$ norms use the uniform physical cube. For $A \subseteq [d]$, write $\chi_A(x) = \prod_{i\in A} x_i$ and $\widehat{F}(A) = \mathbb{E}[\chi_A F]$. Define

$$V(F) = \inf\left\{ \sum_j |c_j| \|a_j\|_2 : F(x) = \sum_j c_j \,\mathrm{ReLU}(a_j \cdot \bar{x}) \text{ on } X \right\}.$$

The same arguments apply to the cost of a specified finite representation, without taking this infimum. The arithmetic–geometric mean inequality gives

$$K_N(\theta) \geq 2\sqrt{d} \sum_j |v_j| \|u_j\|_2 \geq 2\sqrt{d}\, V(F_\theta). \tag{5}$$

Conversely any finite representation can be balanced neuron by neuron so that its energy equals twice $\sqrt{d}$ times its variation cost. Unused neurons may be set to zero.

Partition the physical coordinates into disjoint pairs of rails $(B_i^0, B_i^1)$, with positive even sizes $r_i^0, r_i^1$. Set

$$A_s = \bigcup_{i=1}^m B_i^{s_i}, \quad n_s = |A_s| \quad (s \in \mathbb{F}_2^m), \qquad b(x) = \chi_{A_0}(x).$$

The logical coordinate $z \in G = \mathbb{F}_2^m$ is determined by

$$(-1)^{z_i} = \chi_{B_i^0 \cup B_i^1}(x).$$

Then $\chi_{A_s}(x) = b(x)(-1)^{s\cdot z}$. The logical coordinates are uniform, and the physical base character $b$ is independent of them. The code space $\mathcal{C}$ is the span of the $\chi_{A_s}$. Its orthogonal projection is

$$P_{\mathcal{C}} F(x) = b(x) g(z(x)), \qquad g(z) = \mathbb{E}[b(x)F(x) \mid z(x) = z]. \tag{6}$$

Let $\mathcal{D}$ be the span of all Fourier characters with support contained in at least one $A_s$.

For positive even $n$, define

$$\kappa_n = \frac{\binom{n-2}{(n-2)/2}}{2^{n-1}}, \qquad a(n) = \frac{\sqrt{n}}{\kappa_n}. \tag{7}$$

The alternating binomial identity

$$2^{-n} \sum_{k=0}^n (-1)^k \binom{n}{k} \max(n-2k, 0) = (-1)^{n/2+1} \kappa_n$$

shows that a balanced unit ridge on $n$ coordinates attains absolute parity correlation $\kappa_n/\sqrt{n}$. Signed averaging over its input-coordinate flips represents $\chi_A$ with variation cost $a(|A|)$, using at most $2^{|A|}$ neurons.

**Theorem 2** (Exact unequal-rail norm and stability)**.** *Suppose $m \geq 2$, $R$ is a positive even integer, and all rail sizes are even and lie in $[R, R+\Delta]$, where $\Delta \geq 2$ is an even integer and*

$$R \geq 2^{40} m^6 (\Delta + 1)^4. \tag{8}$$

*For every affine ReLU atom $h_u(x) = \mathrm{ReLU}(u \cdot \bar{x})$,*

$$Q(h_u) := \sum_s a(n_s) |\widehat{h}_u(A_s)| \leq \|u\|_2. \tag{9}$$

*Nonzero equality requires zero bias, exactly one nonzero whole rail per pair, and equal absolute weights on all selected coordinates. Consequently*

$$V\left(\sum_s c_s \chi_{A_s}\right) = \sum_s a(n_s)|c_s|. \tag{10}$$

*Moreover*

$$\|h_u - P_{\mathcal{D}} h_u\|_2^2 \le 2^{16} R^3 \|u\|_2 (\|u\|_2 - Q(h_u)). \tag{11}$$

A complete proof is given in Appendix A. Two features matter in the main argument: the inequality is exact despite unequal rail sizes, and its stability constant is polynomial in $R$, with degree independent of $m$.

For a finite representation with variation cost $B$, weighted Cauchy–Schwarz gives the following consequence. If a code dual $\lambda$ has Fourier coefficients satisfying $|\widehat{\lambda}_s| \le a(n_s)$ and $D = B - \langle \lambda, F \rangle$, then

$$\|F - P_{\mathcal{D}} F\|_2^2 \le 2^{16} R^3 B D. \tag{12}$$

Indeed the weighted sum of the atomic deficits in (9) is at most $D$. This also holds with $B = V(F)$ by approximation.

## 3 Strictly ordered parity costs

Use the rail sizes in Theorem 1: $r_i^0 = R$ and $r_i^1 = R + \delta_i$. Thus

$$n_s = n_0 + \sum_i \delta_i s_i, \qquad n_0 = mR, \qquad w_s = a(n_s), \qquad \widehat{w}(v) = 2^{-m} \sum_s (-1)^{s \cdot v} w_s.$$

The parameter choice satisfies (8), with $\Delta = 8m - 2$.

**Lemma 3** (Walsh signs, distinctness, and a tail bound)**.** *Every nonconstant $\widehat{w}(v)$ is strictly negative, and all the $2^m$ values $\widehat{w}(v)$ are pairwise distinct. If $|v| \ge q \ge 2$, then*

$$|\widehat{w}(v)| \le \frac{31}{mR} \left(\frac{4m}{R}\right)^q. \tag{13}$$

*Proof.* Appendix B proves the exact representation

$$a(r) = \sqrt{2\pi}(r - 3/4) - 2\sqrt{2\pi} \int_0^\infty e^{-rt/2} h(t)\, dt, \qquad h(t) > 0, \quad h(t) \le 3. \tag{14}$$

For $v \ne 0$, its Walsh transform is

$$\begin{aligned} \widehat{w}(v) = & -\mathbf{1}_{v=e_j} \frac{\sqrt{2\pi}\delta_j}{2} \\ & - \frac{2\sqrt{2\pi}}{2^m} \int_0^\infty e^{-n_0 t/2} h(t) \prod_{i:v_i=1} (1 - e^{-\delta_i t/2}) \prod_{i:v_i=0} (1 + e^{-\delta_i t/2})\, dt. \end{aligned}$$

The integral is strictly positive. For $k = |v| \ge 2$, use $1 - e^{-x} \le x$, $1 + e^{-x} \le 2$ and $h \le 3$ to obtain

$$|\widehat{w}(v)| \le \frac{12\sqrt{2\pi}}{n_0} k! \left(\frac{\delta_{\max}}{2n_0}\right)^k \le \frac{31}{mR} \left(\frac{4m}{R}\right)^k.$$

This implies (13).

For distinctness, put $q_i = 2m + 2i - 1$, so $\delta_i = 2q_i$. Every degree $n_s$ lies in an interval of length $6m^2$ starting at $n_0 > (6m^2)^2$. Distinct integers in that interval have distinct squarefree parts: two integers $au^2 < av^2$ with the same squarefree part differ by more than $2\sqrt{n_0}$. Since $a(n)$ is a nonzero rational multiple of $\sqrt{n}$ at even integers, its values at the distinct degrees in this interval are linearly independent over $\mathbb{Q}$.

The coefficients in the Walsh sum, grouped by degree, are the coefficients of

$$P_v(t) = \prod_{i:v_i=1} (1 - t^{2q_i}) \prod_{i:v_i=0} (1 + t^{2q_i}).$$

The cyclotomic polynomial $\Phi_{q_i}$ divides $P_v$ exactly when $v_i = 1$. Indeed the odd integers $q_i \in (2m, 4m)$ divide none of the other $q_j$, and an odd-order root of unity has no power equal to $-1$. Thus different $v$ give different polynomials. The preceding rational independence makes the corresponding Walsh sums different. □

## 4 Calibration that forces a single exceptional fiber

Put $M = 2^m$, $q = \lfloor m/2 \rfloor$, and choose the affine set

$$U = \{z \in G : z_1 = \cdots = z_q = 1, \quad \sum_i z_i = 1 \pmod 2\}. \tag{15}$$

It has size $K = 2^{m-q-1}$ and is a coset $v_0 + H$, where $H$ has codimension $q+1$. Let $J = H + \mathrm{span}(v_0)$ and $E = H^\perp \setminus J^\perp$. Then $|E| = 2^q$, and every character in $E$ equals $-1$ throughout $U$.

The target logical label is

$$t(z) = \begin{cases} +1, & z = 0, \\ -1, & z \neq 0, \end{cases} \qquad f(x) = b(x)t(z(x)).$$

Initially reveal every physical input whose logical coordinate lies outside $U$. Subsequently whole fibers of $U$ will be revealed in a specified order. Let $U_{\mathrm{live}} \subseteq U$ be the nonempty set of fibers not yet revealed. All known labels on a removed fiber are the negative logical label. Write $\mathcal{K} = \{x : z(x) \notin U\}$ for the fixed calibration region. Extra revealed fibers impose additional constraints but are not included in averages with subscript cal.

For $v \in U$, define the star score

$$g_v(z) = -\tfrac{1}{2} + \mathbf{1}_{z=0} + \mathbf{1}_{z=v}, \qquad F_v(x) = b(x)g_v(z(x)).$$

It agrees with the target on every fiber except $v$, and has absolute margin $1/2$ everywhere. Its code coefficients have negative sign at $s = 0$, are nonnegative elsewhere, and vanish on $E$.

Define the code dual through its logical Fourier coefficients:

$$\widehat{\lambda}_s = w_s - 2w_0\mathbf{1}_{s=0} - p_E\mathbf{1}_{s\in E}, \qquad p_E = \frac{2w_0}{|E|}. \tag{16}$$

Its physical version is $b(x)\lambda(z(x))$. Since $w_s \geq w_0$, $|\widehat{\lambda}_s| \leq w_s$, so it is a unit variation-norm dual. Fourier inversion gives

$$\lambda(z) = M\widehat{w}(z) - 2w_0 - 2w_0\mathbf{1}_H(z) + 2w_0\mathbf{1}_U(z). \tag{17}$$

In particular $\lambda(v) = M\widehat{w}(v)$ on $U$, while on the fixed calibration region its sign agrees with $t$ and its magnitude is at least $2w_0$.

The stars saturate the dual and have exact costs

$$V(F_v) = C_v = C_* + \widehat{w}(v), \qquad C_* = \frac{1}{M}\sum_s w_s + \frac{w_0}{2} - \frac{4w_0}{M}. \tag{18}$$

By Lemma 3, these costs are all distinct.

### 4.1 A linear rounding estimate

Consider any finite ReLU score $F$ with absolute margin at least $1/2$ on the full cube and correct margins on all known fibers. Let $B$ be its variation cost, or its variation norm. We only need consider $B \le \max_v C_v$, since the remaining stars are competitors. Write its code projection as $bg$, and set

$$p_v = g(v) + \tfrac{1}{2} \quad (v \in U), \qquad D = B - \langle b\lambda, F\rangle,$$
$$S = \mathbb{E}_{\mathrm{cal}}\big[|\lambda(z)|(fF - \tfrac{1}{2})\big].$$

The expectation with subscript "cal" is the original normalized physical expectation with the indicator of $\mathcal{K}$. Both $D$ and $S$ are nonnegative. A direct decomposition gives

$$B = C_* + D + S + \sum_{v\in U} \widehat{w}(v)p_v. \tag{19}$$

**Lemma 4** (Physical variance control)**.** *For these scores,*

$$L := \|F - P_{\mathcal{C}}F\|_2^2 \le C_{\mathrm{phys}}(D+S), \qquad C_{\mathrm{phys}} = 2^{24}m^2R^4. \tag{20}$$

*Proof.* Let $Z$ be the parity of all physical coordinates. If a proper Fourier subset lies inside a transversal, its complement lies outside every transversal. Therefore, for $A \in \mathcal{D}\cap\mathcal{C}^{\perp}$,

$$\langle A, ZA\rangle = 0, \qquad \|A\|_2^2 = 2\mathbb{E}[\mathbf{1}_{Z=+1}A^2]. \tag{21}$$

The halfcube $Z = +1$ is entirely known because $U$ has odd logical parity.

On a fiber in the fixed calibration region put $e = fF - 1/2 \ge 0$. The signed conditional expectation (6) gives

$$\mathbb{E}_{\mathrm{cal}}|F - P_{\mathcal{C}}F| \le 2\mathbb{E}_{\mathrm{cal}}e \le S/w_0.$$

A variation-cost-$B$ representation satisfies $\|F\|_\infty \le \sqrt{d+1}B$, and $P_{\mathcal{C}}$ is an $L^\infty$ contraction. Hence

$$\mathbb{E}[\mathbf{1}_{Z=+1}(F - P_{\mathcal{C}}F)^2] \le 4\sqrt{d+1}B\mathbb{E}_{\mathrm{cal}}e.$$

Write $E_0 = F - P_{\mathcal{D}}F$ and $A = P_{\mathcal{D}}F - P_{\mathcal{C}}F$. From (12), $\|E_0\|_2^2 \le 2^{16}R^3BD$. Applying (21) and $(u-v)^2 \le 2u^2 + 2v^2$ gives

$$\|A\|_2^2 \le 16\sqrt{d+1}B\mathbb{E}_{\mathrm{cal}}e + 4\|E_0\|_2^2.$$

The two residuals are orthogonal. Consequently

$$L \le B\left[\frac{8\sqrt{d+1}}{w_0}S + 5\cdot 2^{16}R^3D\right].$$

The recurrence for $a(n)$ gives $n \le a(n) \le 3n$. Thus we may use $B \le 5m(R+\Delta) \le 10mR$, $w_0 \ge mR$, and $d+1 \le 4mR$. These bounds imply the looser displayed constant $C_{\mathrm{phys}}$. □

**Lemma 5** (Rounding to a live star)**.** *For some $v \in U_{\mathrm{live}}$,*

$$\|p - e_v\|_1 \le B_{\mathrm{round}}(D+S), \qquad B_{\mathrm{round}} = 2^{31}2^{3m}m^2R^4. \tag{22}$$

*Here $e_v$ is the point mass at $v$, indexed by all of $U$.*

*Proof.* Put $d_z = g(z) - (-1/2 + \mathbf{1}_{z=0})$ outside $U$. Calibration and (17) imply

$$\sum_{z \notin U} |d_z| \le \frac{MS}{2w_0}.$$

The Fourier norm inequality and dual slack give

$$\sum_{s \in E} |\widehat{g}(s)| \le D/p_E, \qquad 2w_s(-\widehat{g}(s))_+ \le D \quad (s \notin E \cup \{0\}).$$

Set $\eta = M(D+S)/p_E$. Because all characters in $E$ equal $-1$ on $U$,

$$\left| \sum_{v \in U} p_v - 1 \right| \le \eta.$$

For any nontrivial character of $H$, choose two extensions to $G$ having opposite signs on $v_0$. They differ by an element of $E$ and both lie outside $H^\perp$. Applying the negative-coefficient inequality to these two extensions bounds the absolute value of the unnormalized Fourier transform of $p$ by $1+\eta$. The zero frequency obeys the same bound. Parseval on the affine cube $U$ gives

$$\sum_{v \in U} p_v^2 \le (1+\eta)^2. \tag{23}$$

Every value of the signed physical score $bF$ has absolute value at least $1/2$. Its conditional mean on a fiber $v$ is $p_v - 1/2$. Therefore

$$V_{\mathrm{frac}} := \sum_{0<p_v<1} p_v(1-p_v) \le ML.$$

Suppose first $\eta \le 1$. Then $\sum p_v^2 - \sum p_v \le 4\eta$, so

$$\sum_{p_v \notin [0,1]} (p_v^2 - p_v) \le 4\eta + V_{\mathrm{frac}}.$$

Round every coordinate to its nearest $b_v \in \{0,1\}$. The distance is at most $p_v^2 - p_v$ outside $[0,1]$ and at most $2p_v(1-p_v)$ inside. Consequently

$$\|p - b\|_1 \le 4\eta + 3V_{\mathrm{frac}}, \qquad \left|\sum b_v - 1\right| \le 5\eta + 3V_{\mathrm{frac}}.$$

If the last bound is less than one, the integer $\sum b_v$ is one. A revealed whole fiber has $p_v \le 0$, so this point mass is live. Otherwise $\eta + V_{\mathrm{frac}} \ge 1/5$; for any live $v$,

$$\|p - e_v\|_1 \le \sqrt{K}\|p\|_2 + 1 \le 2\sqrt{K} + 1 \le (10\sqrt{K}+5)(\eta + V_{\mathrm{frac}}).$$

For $\eta > 1$, the same conclusion follows immediately from (23). Thus in all cases

$$\mathrm{dist}_1(p, \{e_v : v \in U_{\mathrm{live}}\}) \le (10\sqrt{K}+5)M\left(\frac{1}{p_E} + C_{\mathrm{phys}}\right)(D+S).$$

Using $K \le 2^m$, $|E| = 2^q$, $w_0 \ge mR$ and Lemma 4 bounds this constant by $B_{\mathrm{round}}$. □

**Proposition 6** (Every minimum-cost completion is the cheapest star)**.** *At every stage with $U_{\mathrm{live}} \neq \varnothing$, the unique minimum-variation-cost score consistent with all known whole fibers and the closed full-cube margins is $F_v$, where $v$ is the cheapest live fiber in the order* (18)*.*

*Proof.* Let $W = \max_{v \in U} |\widehat{w}(v)|$. For a score no more expensive than a live star, choose the live vertex furnished by Lemma 5. Equations (19) and (22) imply

$$B \geq C_v + (1 - WB_{\mathrm{round}})(D + S).$$

The parameter choice makes $WB_{\mathrm{round}} < 1/2$. Explicitly, Lemma 3 and the displayed rounding constant give

$$\log_2(WB_{\mathrm{round}}) \leq 216 + 3m + 31 \log_2 m - q(58 + 9 \log_2 m) < -1 \quad (m \geq 32).$$

For the last inequality use $q \geq (m-1)/2$; both resulting coefficients are already negative with substantial slack at $m = 32$.

A cheapest star attains equality at its cost. Any other minimizer must therefore have $D = S = 0$. Lemma 4 then gives $L = 0$, and Lemma 5 gives $p = e_v$. Calibration slack zero fixes all remaining code values, so the physical score is exactly $F_v$. The star costs are distinct, proving uniqueness. □

## 5 Finite width, the input sequence, and the posterior

Each star has an exact minimum-variation representation obtained by summing signed parity projections. Its number of neurons is at most

$$\sum_{s \in G} 2^{n_s} \leq 2^{m + n_{\max}} < N_0.$$

At every $N \geq N_0$ it can therefore be zero-padded and balanced to attain energy $2\sqrt{d}\, C_v$. Conversely (5) and Proposition 6 show that every minimum-energy completion has precisely the cheapest live star score.

Order all of $U$ by increasing $C_v$. The input permutation first lists all physical inputs outside $U$. Then it lists the complete physical fibers in this cost order, with any fixed order inside each fiber. Designate the first input in each such fiber. Just before this input is queried, the entire preceding fibers and all fibers outside $U$ have been revealed. The unique minimum-energy score is therefore the star that is positive on this next fiber. The target is negative there. Multiplication by the base character preserves this disagreement.

For completeness, the passage to the actual posterior uses only a finite-dimensional small-noise argument. At a designated round, replace all strict constraints by their closed versions. The feasible set is closed and the quadratic energy is coercive, so minima exist. Requiring the correct sign at the designated point excludes every energy minimizer. It therefore raises the minimum energy by a strictly positive amount. The strict feasible set has the same infimum as its closed version: scale the readout of a minimizing star by $1 + \varepsilon$ and let $\varepsilon \downarrow 0$.

An open ball around a strictly feasible scaled star supplies a denominator whose logarithmic Gaussian rate is arbitrarily close to the minimum. The closed correct-prediction set has a strictly larger rate. Integrating the Gaussian density, or applying its elementary finite-dimensional Laplace principle, shows that its conditional probability tends to zero. This argument sums all parameterizations, including non-minimizing ones; it does not make a tie-breaking assumption.

There are finitely many designated rounds for fixed $m$. Choose the prior scale small enough that their correct-label probabilities sum to less than $\eta$. A union bound then proves the posterior-sampling claim, and probabilities below $1/2$ and $2/3$ prove the other two claims.

It remains to verify the teacher resources. Compute each rail parity, then the $m$ logical parities and their AND; multiplying by the base parity gives $f$. This uses $O(d)$ bounded-fan-in Boolean gates. For a shallow ReLU teacher use the score

$$F_{\text{teacher}}(x) = \sum_{i=1}^{m} \chi_{A_{e_i}}(x) - (m-1)\chi_{A_0}(x) = b(x)\left(\sum_i (-1)^{z_i} - (m-1)\right). \tag{24}$$

It has margin one and the target sign. Parity on $r$ bits is a piecewise-linear function of their sum on the $r+1$ attainable values. Linear interpolation represents it by at most $r+1$ affine ReLU neurons, with incoming coordinate weights one, biases bounded by $r$, and readout coefficients bounded by two. The constant term is a neuron on the constant input. Applying this to the $m+1$ parities in (24) gives width at most $C_m$ and every coefficient at most $C_m$. This completes the proof of Theorem 1.

# A Proof of the exact unequal-rail norm

## A.1 Two elementary analytic tools

**Lemma 7** (Minimum-coordinate parity bound)**.** *For even $n \geq 2$ and any affine ReLU ridge on $n$ Boolean coordinates,*

$$|\widehat{h}([n])| \leq \kappa_n \min_i |w_i|.$$

*Proof.* A coordinatewise nondecreasing $a : \{-1,1\}^k \to [0,1]$ has top Fourier coefficient at most $2^{-k}\binom{k-1}{\lfloor (k-1)/2 \rfloor}$ in absolute value. To see this, average over coordinate permutations and express the resulting nondecreasing sequence on Hamming levels as a convex combination of level thresholds and a constant. The alternating binomial sum of a threshold starting at level $j$ is $(-1)^{k-j}\binom{k-1}{j-1}$.

Input flips make all incoming weights nonnegative. Integrate out a chosen coordinate $i$. The difference $[\mathrm{ReLU}(t+w_i) - \mathrm{ReLU}(t-w_i)]/(2w_i)$ lies in $[0,1]$ and is nondecreasing in the other coordinates. Apply the preceding bound with $k = n-1$. A zero weight gives a zero coefficient directly. □

The affine part of ReLU has no even Fourier coefficient of degree at least two. The identity

$$|z| = \frac{2}{\pi}\int_0^\infty \frac{1-\cos(tz)}{t^2}\,dt$$

therefore gives, up to its fixed degree sign, the coefficient integral

$$\widehat{h}(A) = \frac{(-1)^{|A|/2+1}}{\pi}\int_0^\infty \cos(bt)\prod_{j\in A}\sin(w_j t)\prod_{j\notin A}\cos(w_j t)\,\frac{dt}{t^2}. \tag{25}$$

For the even block supports under consideration these integrals are absolutely convergent.

**Lemma 8** (Block balancing)**.** *For any nonnegative weighted sum of absolute coefficients on nonempty unions of disjoint even blocks, the supremum of its ratio to the augmented incoming Euclidean norm is unchanged by restricting to zero bias and nonnegative weights constant within each block.*

*Proof.* Take absolute values in (25), discard bias and irrelevant-coordinate cosine factors, and flip signs of incoming weights. Call the resulting nonnegative integral $G(w)/\pi$. For block-constant weights it equals the coefficient sum exactly, because every sine and cosine exponent is even.

For a block of size $r$, generalized Hölder, on the disjoint union of coefficient-index copies of the integration measure, gives

$$G(w) \le \prod_{j \text{ in block}} G(w \text{ with this whole block replaced by } w_j)^{1/r}.$$

Iterate over blocks. If $V$ replaces each block by an independent uniformly sampled coordinate magnitude from that block, the result is $G(w)/\pi \le \exp(\mathbb{E}\log Q(h_V))$. If $M_*$ is the restricted supremum, this is at most $M_* \exp(\mathbb{E}\log\|V\|) \le M_*\sqrt{\mathbb{E}\|V\|^2} = M_*\|w\|$. Bias and irrelevant weights only increase the original norm. □

### A.2 Localization of a possible violating balanced atom

By Lemma 8, first consider balanced nonnegative rails. Within each pair call their magnitudes $a_i \ge b_i \ge 0$, exchanging rail names if necessary. Normalize the physical norm to one. Put $n = mR$ and

$$\alpha = \frac{a(m(R+\Delta))}{a(mR)} \le 1 + 2\Delta/R.$$

The bound follows from $a(n+2)/a(n) = \sqrt{n(n+2)}/(n-1)$, which makes $a(n)/(n-1)$ decreasing along even integers. Reducing every even sine/cosine exponent to $R$ increases the nonnegative integrands. Hence $Q_r \le \alpha Q_R$.

Clarkson's elementary inequality gives, for even $R$,

$$\sin^R(at)\cos^R(bt) + \cos^R(at)\sin^R(bt) \le \tfrac{1}{2}\{\sin^R((a+b)t) + \sin^R((a-b)t)\}.$$

Multiply over rail pairs, integrate, and apply Lemma 7. With independent symmetric signs $\varepsilon_i$,

$$Q_R \le \sqrt{n}\,\mathbb{E}\min_i(a_i + \varepsilon_i b_i). \tag{26}$$

Suppose $Q_r \ge 1$, and put $h = (\alpha\sqrt{n})^{-1}$ and $K_0 = 3m\Delta/R$. The expectation in (26) is at least $h$ and at most every $a_i$. Since $R\sum_i a_i^2 \le 1$,

$$h \le a_i \le h(1+K_0).$$

For $i \ne j$, concavity of the minimum gives

$$\mathbb{E}\min_k(a_k + \varepsilon_k b_k) \le \frac{a_i + a_j - \max(|a_i - a_j|, b_i)}{2},$$

so $b_i \le 2hK_0$.

Let $A$ be the selected transversal, $D_0 = |A|$, and let $a$ be the weighted mean of the selected rail magnitudes. Write the selected weights as $a + \delta_j$, with $\sum_A \delta_j = 0$, and denote the unused weights by $u$. Set $\tau^2 = \|\delta\|^2 + \|u\|^2$. The preceding bounds imply

$$\tau/a \le 10m^{3/2}\Delta/\sqrt{R}. \tag{27}$$

### A.3 A uniform cusp near a selected parity

We first record a useful gap. Let $U = \sum_i c_i \sum_{j\in B_i} Y_j$ be a sum of independent even blocks, each of size at least four and with a constant coefficient inside a block. For every offset $b$,

$$\sum_{\varnothing\neq I} \left| \widehat{|b+U|}\Big(\bigcup_{i\in I} B_i\Big) \right| \le \tfrac{1}{2}\mathbb{E}|U|. \tag{28}$$

Indeed (25) bounds the coefficients by those at $b = 0$. Condition each block parity to be $(-1)^{|B_i|/2}$. The projected value on this fiber equals the mean of $|U|$ minus the sum of the absolute nonconstant block coefficients. This conditional distribution keeps the mean zero and the variance unchanged. Its fourth moment is at most four times the square of its variance: for a size-four block this follows by direct expansion, and for larger blocks all fourth moments are unchanged. Independence preserves the bound. Hölder therefore gives $\mathbb{E}_{\text{fiber}}|U| \ge \frac{1}{2}\sqrt{\mathbb{E}U^2} \ge \frac{1}{2}\mathbb{E}|U|$, proving (28).

Write $S_0 = \sum_{j\in A} X_j$ and $T = \delta\cdot X + u\cdot Y$. Away from $S_0 = 0$,

$$\mathrm{ReLU}(aS_0 + T) = \mathrm{ReLU}(aS_0) + \mathbf{1}_{S_0>0}T + \mathcal{R}, \qquad |\mathcal{R}| \le |T|\mathbf{1}_{|T|\ge 2a}.$$

At $S_0 = 0$ add the exact term $\mathrm{ReLU}(T)$. If $\tau \le a$, the Rademacher subgaussian bound implies

$$\mathbb{E}|\mathcal{R}| \le 3\tau e^{-a^2/\tau^2}. \tag{29}$$

Let $p_0 = \mathbb{P}(S_0 = 0)$ and $\epsilon_0 = (-1)^{D_0/2+1}$. The selected coefficient satisfies

$$\epsilon_0\widehat{h}(A) = \kappa_{D_0}a - \frac{p_0}{2}\mathbb{E}[|T| \mid S_0 = 0] + \epsilon_0\widehat{\mathcal{R}}(A).$$

The regular linear term vanishes because $\sum_A \delta_j = 0$. On the balanced slice the selected parity is $-\epsilon_0$. Every other transversal coefficient comes only from the equator term and the remainder. Apply (28) to the unused blocks, conditionally on the selected coordinates. Since $\mathbb{E}|b+U| \ge \mathbb{E}|U|$, summing those coefficients and using $a_{\max}/a(D_0) \le \alpha \le 3/2$ gives

$$Q_r \le \sqrt{D_0}a - \frac{a(D_0)p_0}{8}\mathbb{E}[|T| \mid S_0 = 0] + 2^m a_{\max}\mathbb{E}|\mathcal{R}|. \tag{30}$$

The selected coefficient keeps its sign in this neighborhood; this also follows directly from $\tau/a \ll 1$ and $p_0/(2\kappa_{D_0}) = (D_0-1)/D_0$.

For a zero-sum vector $\delta$ on a balanced slice,

$$\mathbb{E}[(\delta\cdot X)^2 \mid S_0 = 0] = \frac{D_0}{D_0-1}\|\delta\|^2,$$

$$\mathbb{E}[(\delta\cdot X)^4 \mid S_0 = 0] = \frac{3D_0(D_0-2)}{(D_0-1)(D_0-3)}\|\delta\|^4 - \frac{2D_0^2}{(D_0-1)(D_0-3)}\sum_j \delta_j^4 \le 12\|\delta\|^4.$$

These identities follow by expanding and using the two- and four-point moments of the uniform balanced slice. Adding the independent outside sum gives conditional second moment at least $\tau^2$ and fourth moment at most $12\tau^4$. Thus

$$\mathbb{E}[|T| \mid S_0 = 0] \ge \tau/\sqrt{12}, \qquad p_0 \ge 1/\sqrt{2D_0}.$$

The cusp loss in (30) is at least $a(D_0)\tau/(8\sqrt{24D_0})$. In contrast, the remainder term is at most $3\cdot 2^m a_{\max}\tau e^{-a^2/\tau^2}$. By (27) and (8),

$$\frac{a^2}{\tau^2} \ge \frac{R}{100m^3\Delta^2} > \log(72\cdot 2^m\sqrt{24D_0}).$$

For an explicit check, $D_0 \le 2mR$ bounds the logarithm by $7 + 2m + \frac{1}{2}\log R \le 2\sqrt{R}$, whereas the left side is at least $(2^{20}/100)\sqrt{R}$. Thus if $\tau > 0$,

$$Q_r < \sqrt{D_0}a \le \sqrt{D_0 a^2 + \tau^2} = 1,$$

a contradiction. The case $\tau = 0$ is precisely a balanced selected parity and attains equality.

For arbitrary incoming weights, equality in block balancing forces every independently sampled balanced vector to have this form. A partially zero rail, two nonzero opposite rails, or unequal selected coordinate magnitudes would produce a positive-probability strict sample. Thus the same equality classification holds before balancing, and bias and irrelevant weights must vanish. This proves (9); signed parity averaging gives (10).

### A.4 Polynomial squared-distance stability

Let $\mathcal{A}$ be the union of balanced selected-transversal parameter rays, including zero and all input sign choices. For balanced physical vectors $v$, the preceding localization and cusp proof gives

$$\operatorname{dist}(v, \mathcal{A})^2 \le 100R\|v\|(\|v\| - Q(h_v)). \tag{31}$$

Here are the constants. Normalize $\|v\| = 1$ and put $\varepsilon = 1 - Q(h_v)$. If $\varepsilon \ge \Delta/(100R)$, use the zero ray. Otherwise replace $\alpha$ above by $\alpha/(1-\varepsilon) \le 1 + 3\Delta/R$ and $K_0$ by $4m\Delta/R$. Then $\tau/a \le 20m^{3/2}\Delta/\sqrt{R}$. The same cusp comparison, with the denominator 100 replaced by 400, gives $Q(h_v) \le \sqrt{D_0}a$, hence $\operatorname{dist}(v, \mathcal{A})^2 \le \tau^2 \le 2\varepsilon$.

We now transfer this bound before block balancing without an exponentially small sampling probability. Assume nonnegative relevant weights and independently sample one coordinate magnitude $U_i^j$ from each rail; let $V$ repeat this value throughout the rail. Put $a_i = \max(U_i^0, U_i^1)$, $b_i = \min(U_i^0, U_i^1)$, and $\bar{a} = m^{-1}\sum_i a_i$. The squared scalar distance to selecting one rail of common magnitude is

$$d_0(U)^2 = \sum_i b_i^2 + \sum_i (a_i - \bar{a})^2.$$

Write $\eta = \mathbb{E}d_0(U)^2$ and $r_{\max} = \max r_i^j \le 2R$. For independent rail samples $X, Y$ with $A = \mathbb{E}X^2 \ge B = \mathbb{E}Y^2$ and $A + B > 0$ (the zero case is immediate),

$$\mathbb{E}\min(X^2, Y^2) \ge \frac{\mathbb{E}X^2Y^2}{r_{\max}(A+B)} \ge \frac{B}{2r_{\max}}.$$

Choose deterministically in every pair the rail of larger second moment. Call its sample $Z_i$ and the discarded sample $T_i$. Then $\mathbb{E}\sum T_i^2 \le 2r_{\max}\eta$ and $|Z_i - a_i| \le T_i$. If $P$ projects off the constant vector in $\mathbb{R}^m$, independence of the $Z_i$ gives

$$\min_\lambda \mathbb{E}\sum_i (Z_i - \lambda)^2 \le \frac{m}{m-1}\mathbb{E}\|PZ\|^2 \le (4 + 8r_{\max})\eta.$$

Adding discarded rails and restoring physical rail-size weights yields

$$\operatorname{dist}(w, \mathcal{A})^2 \le 24r_{\max}\mathbb{E}\operatorname{dist}(V, \mathcal{A})^2. \tag{32}$$

Normalize the original augmented vector to one and again put $\varepsilon = 1 - Q(h_u)$. For $\varepsilon < 1/2$, the block-balancing geometric-mean inequality and $\mathbb{E}\|V\|^2 \le 1$ imply

$$\mathbb{E}\log\frac{\|V\|}{Q(h_V)} \le -\log(1-\varepsilon) \le 2\varepsilon, \qquad \mathbb{E}\left[1 - \frac{Q(h_V)}{\|V\|}\right] \le 2\varepsilon.$$

Also $\|V\|^2 \le r_{\max}$ pointwise. Equations (31) and (32) therefore give

$$\operatorname{dist}(w, \mathcal{A})^2 \le 48(100R) r_{\max}^2 \varepsilon.$$

The bias and irrelevant squared norm are at most $2\varepsilon$, since $Q(h_u)$ is bounded by the relevant norm. Altogether

$$\operatorname{dist}(u, \mathcal{A})^2 \le 2^{16} R^3 \varepsilon.$$

For $\varepsilon \ge 1/2$, the zero ray suffices. Homogeneity proves the corresponding statement at arbitrary norm. Finally ReLU is 1-Lipschitz and $\mathbb{E}[\bar{x}\bar{x}^\top] = I$, so parameter squared distance bounds function squared $L^2$ distance. Every face atom lies in $\mathcal{D}$, proving (11).

# B An exact Bernstein representation

The gamma-function identity for (7) is

$$a(r) = 2\sqrt{\pi r}\frac{\Gamma(r/2)}{\Gamma((r-1)/2)} = 2\sqrt{2\pi}\, A(r/2), \qquad A(z) = \sqrt{z}\frac{\Gamma(z)}{\Gamma(z-1/2)}.$$

Define

$$k(t) = \sqrt{\frac{t}{1-e^{-t}}}, \qquad B(t) = \frac{1}{\pi}\int_0^1 \frac{k(tu)}{\sqrt{u(1-u)}}\, du, \qquad h(t) = \tfrac{1}{2}B'(t) - B''(t).$$

The limits at zero are $k(0) = B(0) = 1$ and $B'(0) = 1/8$. For $t > 0$, let

$$D(t) = \frac{1}{t} - \frac{1}{e^t - 1}.$$

Then $0 < D(t) < 1/2$ and $D'(t) = -1/t^2 + 1/[4\sinh^2(t/2)] < 0$. Since

$$k'/k = D/2, \qquad k''/k = D^2/4 + D'/2 < D/8,$$

we have $B' > 0$ and $B'' < B'/4$. In particular $h > 0$.

Write $\mathcal{L}$ for the Laplace transform and $b_0(t) = (1-e^{-t})^{-1/2} = t^{-1/2}k(t)$. The beta integral and convolution give

$$\frac{\Gamma(z)}{\Gamma(z+1/2)} = \frac{1}{\sqrt{\pi}}\mathcal{L}[b_0](z), \qquad B = \frac{1}{\pi}(b_0 * t^{-1/2}), \qquad z\mathcal{L}[B](z) = 1 + \mathcal{L}[B'](z).$$

Therefore integration by parts yields

$$\begin{aligned} A(z) &= (z-1/2)z\mathcal{L}[B](z) \\ &= z - 3/8 + \mathcal{L}[B'' - B'/2](z) = z - 3/8 - \int_0^\infty e^{-zt} h(t)\, dt. \end{aligned}$$

All functions and derivatives have at most polynomial growth, so these integrals converge for $z > 0$. At $z = 1/2$, $A$ extends continuously to zero. Consequently

$$A(x + 1/2) = x + \int_0^\infty e^{-t/2}(1 - e^{-xt})h(t)\, dt \quad (x \ge 0),$$

which is the claimed strict Bernstein representation.

For the crude bound $h \le 3$, use $k(t) \le \sqrt{1+t}$ and $D(t) \le \min(1/2, 1/t)$. For $0 < t \le 1$,

$$|D'(t)| = \frac{2(\cosh t - 1) - t^2}{4t^2 \sinh^2(t/2)} \le 2(\cosh 1 - 1) - 1 < 1;$$

the numerator's power series starts at degree four and the denominator is at least $t^4$. For $t \ge 1$, $|D'(t)| \le t^{-2}$. The displayed derivative identities then give $k' \le 1$ and $|k''| \le 2$ on the whole positive line. Hence $B' \le 1$, $|B''| \le 2$, and $0 < h \le 5/2 < 3$. This proves (14) with all bounds used above.